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Parameter Identification

Source and modifications. This manual is an adaptation of two Quanser student workbooks for the QUBE-Servo 2 — *Friction Identification* (QUBE-SERVO 2 Workbook – Student, v1.2, © 2021 Quanser Inc.) and *Parameter Estimation* (QUBE-SERVO 2 Workbook – Student, v1.0, © 2020 Quanser Inc.) — combined into a single laboratory experiment and used under the licence reproduced at the end of this document.

Retained from the original workbooks: the background on the nonlinearities in a motion control system — amplifier saturation, encoder quantization and friction — together with the Coulomb and viscous friction model, Figures 1 to 3 and Equations (1) to (9); the Coulomb and viscous friction identification procedure in Section 6.1; the friction model validation exercise in Section 6.2; and the motor resistance and back-emf identification procedure with Tables 2 and 3 in Section 6.3. The armature schematic and parameter table of the *Parameter Estimation* workbook are not reproduced here; this manual refers back to Table 1 of the Lab 2 manual instead.

Modified from the original: the supplied Simulink models have been removed, so you build the models yourself from your Lab 2 work; the parameter script qube2_param.m is provided unchanged, with a short addition for the encoder conversion factors; and the identification procedures have been rewritten with specific test voltages, a MATLAB Live Script requirement for the curve fit, and an added safety note on stalling the motor by hand.

Written for ENME 561/661 by the course instructor: the materials list, end-of-session shutdown procedure, objective and deliverables; the requirement to design your own Simulink model for the parameter measurements and explain what it measures; the transfer function formed from the experimentally identified values and the step-response comparison against the Lab 2 model; the peer design review of the Lab 2 experiment design in Section 7, including Table 4; the closing comparison of your own Lab 2 experiment design with the procedure provided here; and the self-assessment question.

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1 Materials

- Hardware: QUBE-Servo 2, Cables, Laptop
- **Software:** Stall detection block and the `qube2_param.m` parameter script (available on D2L)
- QUBE-Servo 2 User Manual.pdf (available on D2L)

2 End-of-Session Shutdown Procedure

At the end of each lab session, please ensure the following steps are completed:

1. Make sure the QUARC controller is stopped.
2. Turn off the QUBE-Servo 2.
3. Shut down the laptop.
4. Place the laptop in the designated cabinet.

3 Objective

In this lab you extend the DC motor model you built in Lab 2 to account for the nonlinearities a linear model ignores: amplifier saturation and friction. You identify the Coulomb friction voltage and the viscous friction coefficient from your own measurements, add them to your model, and check the result against the real motor. You then identify the motor resistance and the back-emf constant experimentally, form a transfer function from the values you measured, and compare it with the one you derived in Lab 2. Finally, you review another team's Lab 2 experiment design.

4 Deliverables

- **Section on Friction Identification**: Submit one **.mlx file** and its corresponding results in a **PDF**.
- **Section on Model Friction Validation**: Submit one **.slx file** with the **results and discussion** included in the **main PDF**.
- **Section on Motor Resistance and back-emf Constant Identification**: Submit the **results and discussion** included in the **main PDF**.
- **Printed Lab 2 design:** bring a paper copy of your Lab 2 Section 7 experiment design to the first Lab 3 session and hand it to your teaching assistant.
- **Peer review:** submitted through the **Microsoft Form** during the first Lab 3 session (Section 7). Nothing is uploaded to D2L for this item.

In total: Two **PDF files**, one **.mlx file**, and one **.slx file**.

Answering the questions. Each question is tagged with the form its answer must take — *[number]*, *[figure]*, *[screenshot]*, *[table]* or a stated number of sentences. Steps tagged *[no write-up needed]* are there to be done, not described; the file or figure you submit is the evidence. Do not restate the procedure.

Submission checklist. Everything that is graded for Lab 3 is listed here. Tick each row as you complete it.

✓	What to submit	Where	When
	Main report PDF (Sections 6.1 to 6.3, including Table 1)	D2L Dropbox	Lab 3 deadline
	MATLAB Live Script, .mlx (Section 6.1)	D2L Dropbox	Lab 3 deadline
	PDF export of the Live Script	D2L Dropbox	Lab 3 deadline
	Simulink model, .slx (Sections 6.1 and 6.2)	D2L Dropbox	Lab 3 deadline
	Lab 2 Section 7 experiment design, printed copy	Hand to your TA	Start of the 1st Lab 3 session

✓	What to submit	Where	When
	Peer review of another team's design	Microsoft Form	During the 1st Lab 3 session
	Lab 3 exit survey (individual)	Microsoft Form	Lab 3 deadline

Before you write anything else, fill in Table 1. It is the first thing your teaching assistant will read, and every value in it must also appear in the body of the report with its supporting figure or table.

Table 1: Results summary

Quantity	Your value	Where the evidence is
Positive Coulomb friction voltage Vc+ (V)		
Negative Coulomb friction voltage Vc− (V)		
Total viscous coefficient from the fit (V·s/rad)		
Viscous friction coefficient μf (V·s/rad)		
Average motor resistance Rm (Ω)		
Average back-emf constant km (V·s/rad)		
Experimental transfer function Ω(s)/Vm(s)		

5 Background

5.1 Nonlinearities in Modelling

Linear models are often used to characterize the motion of DC servo motors, along with many other systems. While they can be adequate for many applications, it is important to know what nonlinearities are neglected during the modelling process and how they can affect the system response. The major nonlinearities in a motion control system (e.g. servo) are:

- saturation of the motor amplifier, and
- friction in the motor.

The amplifier that drives the motor has a power supply of 10 V. This means that the voltage from the amplifier can never exceed V_{max} = 10 V. A consequence is that the current through the motor is also limited. This limitation implies that the motor transfer function without considering saturation does not describe the system well for large signals ($u(t)$).

The other main nonlinearities are due to friction. The block diagram of the DC motor is shown in Figure 1. It incorporates the amplifier limitation and the friction nonlinearities.

Rotary and linear motion control systems based on mechanical systems often have friction. The Coulomb (static) and viscous (dynamic) friction model is shown in Figure 2. This is one of the most common friction models. This model includes a discontinuity at zero and a linear gain for all other values.

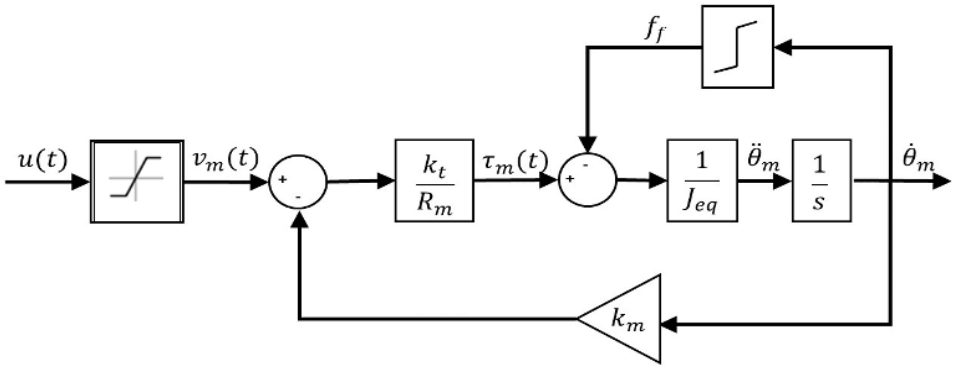


Figure 1 DC motor model with amplifier limit and Coulomb and viscous friction

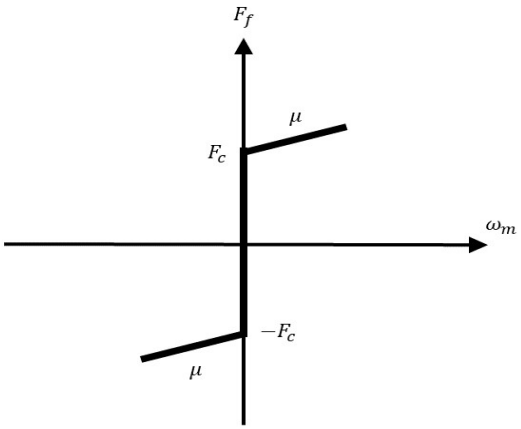


Figure 2 Coulomb and viscous friction model (Torque vs angular velocity)

Given the angular velocity of the motor, ω_m, friction can be modelled as follows:

$$F_f = \mu\omega_m + F_c sgn(\omega_m), \qquad (1)$$

where μ is the viscous friction coefficient and F_c is the Coulomb friction parameter, and

$$sgn(x) = \begin{cases} -1 \; if \; x < 0 \\ 0 \; if \; x = 0 \\ +1 \; if \; x > 0 \end{cases}.$$

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Please note that Equation 1 is in torque units. Viscous friction models the dynamic friction, and Coulomb friction models the static friction. The force needed to overcome the static friction is known as the *breakaway force* (F_c). Any larger force results in acceleration of the shaft.

The motor shaft equation can be modelled as follows:

$$J_{eq}\dot{\omega}_m(t) = \tau_m(t), \qquad (2)$$

where J_{eq} is the total moment of inertia acting on the motor shaft and τ_m is the applied torque from the DC motor. Including friction, this becomes

$$J_{eq}\dot{\omega}_m(t) = \tau_m(t) - F_f, \qquad (3)$$

where F_f represents the torque effects from friction. Based on the current applied, the torque is

$$\tau_m = k_t i_m(t), \quad (4)$$

where k_t is the torque constant. Using Kirchoff's Voltage Law and the assumption that the motor inductance is much less than its resistance, the motor current (i_m) is given by

$$i_m(t) = \frac{v_m(t) - e_b}{R_m} \quad , (5)$$

where v_m is the applied voltage, and R_m is the terminal resistance, and e_b is the *back-emf voltage* and can be modelled as

$$e_b = k_m \omega_m(t) \ , (6)$$

where k_m is the motor back-emf constant.

Applying Equation 1 and Equations 4-6 to Equation 3, the motion equation becomes

$$J_{eq}\dot{\omega}_m(t) = \frac{k_t}{R_m}\left(v_m(t) - k_m\omega_m(t)\right) - \mu\,\omega_m(t) - F_c sgn(\omega_m). (7)$$

Since back-emf voltage effectively behaves as viscous friction, we can lump together back-emf and the effects of friction and redefine it in terms of voltage such that

$$J_{eq}\dot{\omega}_m(t) = \frac{k_t}{R_m}\left(v_m(t)\right) - F_f' \qquad (8)$$

where

$$F_f' = \left(k_m \frac{k_t}{R_m} + \mu\right)\omega_m(t) + F_c sgn(\omega_m). \qquad (9)$$

Equation 9 is similar to Equations 1. Thus, the coefficient of $\omega_m(t)$ is in Equation 9 is called the equivalent viscous damping $\left(k_m \frac{k_t}{R_m} + \mu\right)$ which includes the back-emf parameter, k_m, and the viscous friction μ of the motor. The Coulomb friction term F_c is with respect to the torque.

From the torque form to the voltage form.

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Equation (8) defines F_f' in terms of **torque**, but in Section 6.1 you apply a **voltage** and measure a speed. The two are related by the factor R_m/k_t. Multiplying Equation (8) through by R_m/k_t gives

$$\frac{R_m}{k_t} J_{eq}\, \dot{\omega}_m = v_m - \frac{R_m}{k_t} F_f' \qquad (10)$$

and substituting Equation (9) for F_f',

$$\frac{R_m}{k_t} J_{eq}\, \dot{\omega}_m = v_m - \frac{R_m}{k_t}\left(k_m \frac{k_t}{R_m} + \mu\right)\omega_m - \frac{R_m}{k_t} F_c\, \mathrm{sgn}(\omega_m) \qquad (11)$$

$$\frac{R_m}{k_t} J_{eq}\, \dot{\omega}_m = v_m - \left(k_m + \frac{R_m \mu}{k_t}\right)\omega_m - \left(\frac{R_m}{k_t} F_c\right)\mathrm{sgn}(\omega_m). \qquad (12)$$

Define the **viscous friction coefficient referred to voltage** and the **Coulomb friction voltage** as

$$\mu_f = \frac{R_m}{k_t}\mu, \qquad V_c = \frac{R_m}{k_t} F_c.$$

The coefficient of ω_m in Equation (12), $(k_m + \mu_f)$, is the **equivalent viscous damping**: it contains the back-emf constant as well as the viscous friction of the motor. At steady state $\dot{\omega}_m = 0$, so Equation (12) becomes

$$v_m = (k_m + \mu_f)\omega_m + V_c\, \mathrm{sgn}(\omega_m). \qquad (13)$$

Equation (13) is the line you fit in Section 6.1, with the measured speed on the horizontal axis and the applied voltage on the vertical axis. Read it carefully:

- the **slope** of the fitted line is $k_m + \mu_f$, **not** μ;
- the **intercept** is V_c, **not** F_c;
- both are expressed with respect to voltage, so their units are V·s/rad and V, not N·m·s/rad and N·m.

The fit therefore gives the *sum* $k_m + \mu_f$, not either parameter on its own. Once k_m has been identified independently in Section 6.3, the viscous friction coefficient follows as $\mu_f = \text{slope} - k_m$.

6 In-Lab Exercises

Before you build or run any model in this lab, run `qube2_param.m` from the MATLAB command window. It defines the QUBE-Servo 2 datasheet parameters as MATLAB variables. Use those variable names in your Gain blocks instead of typing numbers — the model stays readable, and a parameter can be changed in one place rather than through the diagram.

The script already computes the equivalent moment of inertia Jeq from the hub and disc parameters; check that it agrees with the value you derived in Lab 2. Add the friction parameters to the same script once you have identified them in Section 6.1.

6.1 Friction Identification

In this lab, you need to create a Simulink code that applies different voltages to the motor and outputs the corresponding angular velocity. This is used to identify the torque needed to overcome the static friction to get the Coulomb friction parameter, as well as find the viscous friction coefficient by looking at the changes in motor velocity with respect to the torque applied.

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1. Create a Simulink code that can achieve the needs described above. For this system, the input is best set up as a constant voltage followed by a **Slider Gain block** to apply a voltage with a range of ± 5. *[no write-up needed]*

   Note: the value of the Slider Gain block can be changed while the system is running.

Note. The model you build here contains no simulated motor — the motor is the real hardware. Start from the filtering model you built in Lab 2: it already applies a voltage to the motor and reads shaft speed through the low-pass filter. Replace the square-wave source with a Constant block followed by a Slider Gain with a range of ±5 V, so the applied voltage can be changed while the model is running, and keep the stall-detection block connected.

2. Build and run the Simulink code. *[no write-up needed]*

3. To find the Coulomb friction in the DC motor of the QUBE-Servo 2, gradually increase the voltage applied to the motor in small increments, e.g. $0.01V$, until you observe a motion in the inertial disc. Once the inertial wheel starts to rotate (i.e. $\omega_m(t) > 0$), it means you have overcome the static friction. The maximum voltage that still keeps the inertial wheel/DC motor at rest is the positive Coulomb friction voltage $\boldsymbol{V_{c+}}$. Report the value of the positive Coulomb friction voltage. Add the plot for the shaft speed versus voltage. *[number + figure]*

4. To identify the negative Coulomb friction voltage, $\boldsymbol{V_{c-}}$ start at 0V and repeat the procedure by decreasing the voltage applied to the motor in small increments (e.g. 0.01V). Report the value of the negative Coulomb friction voltage. Add the plot for the shaft speed versus applied voltage, which shows the negative Coulomb friction voltage *[number + figure]*

5. To identify the viscous friction, vary the voltage between -4V and 4V in increments of 1V. Save the voltage (in Volts) and the resulting angular velocity (in rad/s) in a file that you can later open with an mlx file. *[no write-up needed]*

6. Create a Live Script (mlx file) in MATLAB to generate a plot, where the x-axis is the measured velocity and the y-axis is the applied voltage. *[figure]*

7. Fit a polynomial to the data using the MATLAB polyfit[1] function, and report the polynomial as a function of the shaft speed. Then, evaluate the polynomial at each point using the MATLAB polyval[2] function. This is to create a linear interpolation of the measured data and display its corresponding equation. Plot estimated values and measured values in one figure and compare them. *[figure + polynomial]*

   Hint: In order to have a good fit, you need to separate the data into two sets ($-V_c$ to -4) and ($+V_c$ to 4) and fit two functions to these data.

   Hint: Make sure you use at least 3 significant digits.

   Optional: You may watch the MATLAB Curve Fitting Onramp:

   https://matlabacademy.mathworks.com/details/curve-fitting-onramp/orcf

[1] https://in.mathworks.com/help/matlab/ref/polyfit.html
[2] https://www.mathworks.com/help/matlab/ref/polyval.html

8. Report the value of the coefficient that you identified. Explain what this value is exactly. *[number + 3-5 sentences]*

9. Stop the QUARC controller. *[no write-up needed]*

6.2 Model Friction Validation

1. Modify your Simulink code from the block diagram Section of Lab 2, to include the effect of Coulomb friction as found in this lab. This can be done with the **Coulomb & Viscous Friction block** in Simulink. Add this modified block diagram to your Simulink code so it can be run in parallel with the motor. Ensure that your setup includes: *[no write-up needed]*

   - The original motor model without friction,
   - The modified motor model with Coulomb friction, and
   - The hardware system,

   all running together for comparison.

2. **Report** the values that you select for the Coulomb & Viscous Friction block in Simulink, and explain how you select them. *[values + 2-3 sentences]*

   When defining the viscous friction in the Coulomb & Viscous Friction block, make sure to use the viscous coefficient which is the one that includes both the back-emf and friction effects (i.e. the total friction).

   The Coulomb friction model assumes the positive and negative force, F_c, to be symmetrical. If the positive and negative Coulomb friction found are not the same, i.e. $V_{c+} \neq V_{c-}$, take the largest one and let that be the equivalent Coulomb friction, i.e. $V_c = \max\{V_{c+}, V_{c-}\}$.

3. Set the code to apply an input voltage that is a **square wave** going from -1V to 1V at 0.4 Hz. *[no write-up needed]*

4. Build and run the code in QUARC. *[no write-up needed]*

5. Show the output of the block diagram with friction, without friction and the physical motor position in one figure. *[figure]*

6. Does including friction improve the accuracy of the model? Explain your results and how you obtained the conclusion that you did. *[2-3 sentences]*

7. Stop the QUARC controller. *[no write-up needed]*

6.3 Motor Resistance and back-emf Constant Identification

In this section, you need to estimate motor resistance (R_m) and Motor back-emf constant (k_m).

Note: **Configure the HIL Read block** to also read from the **current** sensor on analog input **channel #0**.

1. The first parameter you need to find for the system is the motor resistance (R_m), which can be found by applying constant voltages to the motor and measuring the corresponding current *while holding the motor*

shaft stationary. **Derive an expression** that will allow you to solve for R_m under these conditions. Explain how you derived this and what assumptions you considered. *[equation + assumptions]*

2. The second parameter you need to find is the motor back-emf constant, k_m, which can be found by applying **constant voltage**s to the motor and measuring both corresponding steady-state current and speed (in radians per second). Assuming that the motor resistance is known, derive an expression that will allow you to solve for k_m. Explain how you derived this and what assumptions you considered. *[equation + assumptions]*

3. Connect and power up the QUBE-Servo 2 system and ensure the inertial disc load is mounted. *[no write-up needed]*

4. Design a Simulink code that can be used to find the required values for this experiment. Add a screenshot of the model to your report, and state in one sentence what it lets you measure and why that gives you the parameter. Do not describe how you assembled it. *[screenshot + 1 sentence]*

5. To experimentally estimate the motor resistance, apply a set of constant voltages(-5,-4,-3,-2,-1,1,2,3,4,5 volts) for a couple of seconds to the QUBE-Servo 2 using a Simulink code that can measure the current in the system while a voltage is applied. For each measurement, *hold the motor shaft stationary* by grasping the inertial disc load to stall the motor and record the current measurement displayed in the *Current (A)* display. Fill the following table with the measured current for different voltages and calculate the corresponding resistance. *[table]*

Note: Be very careful when stalling the motor to avoid injuring yourself or damaging the motor.

Table 2. Recorded data for the identification of motor resistance

Applied Voltage: V_m (V)	**Measured Current: I_m (A)**	**Resistance: R_m (Ω)**
-5		
-4		
-3		
-2		
-1		
1		
2		
3		
4		

5		

6. Take the average of all the measured resistance values and compare this with the motor resistance value from the specification sheet in the datasheet provided in Lab 2. How close was your experimental value? *[number + 1 sentence]*

7. To experimentally estimate the motor back-emf constant, repeat the same procedure by applying a set of constant voltages to the QUBE-Servo 2 with the motor free to spin (i.e. do not stall the motor) and record the measured speed and current in the following Table. *[table]*

Table 3. Recorded data for identification of motor back-emf constant

V_m (V)	Measured Speed: ω_m (rad/s)	Measured Current: I_m (A)	Back-emf: k_m (V·s/rad)
-5			
-4			
-3			
-2			
-1			
1			
2			
3			
4			
5			

7. Take the average of the measured back-emf values and compare this with the motor back-emf value from the specification sheet in the datasheet provided in Lab 2. How close was your experimentally found value? What are the reasons for any discrepancy? *[number + 1-2 sentences]*

8. Formulate the **transfer function for Voltage to Speed** based on your experimentally found values. How does this function compare to the transfer function you found in Lab 2? What is the difference when the step response for both models is plotted in MATLAB? Provide the two plots in one figure. *[expression + figure + 2-3 sentences]*

Hint: you may use the MATLAB *tf* function[3] and the MATLAB *step* function[4].

9. Stop QUARC. *[no write-up needed]*

10. In Lab 2, you designed your own experiments to identify the motor resistance (R_m) and motor back-emf constant (k_m). In Lab 3, you were provided with specific instructions to conduct experiments for identifying the same parameters. Compare your Lab 2 design with the provided instructions. Discuss whether your Lab 2 experiment design is applicable. If it is, conduct your original experiment, and report the identified motor resistance and motor back-emf constant. Are these values different from those obtained using the Lab 3 instructions? Discuss your result. If the designed experiment is not feasible, explain why it is not feasible. *[4-6 sentences]*

If you were to grade yourself, what grade would you give out of 10? *[one line]*

7 Peer Design Review

At the beginning of the first lab session, hand your printed Lab 2 experiment design to your teaching assistant. Write your first names and your group number from D2L at the top of the page. Your teaching assistant will pass your design to a different group for review and will give you another group's design in exchange.

Spend about 25 minutes reading the design you receive and evaluating it against the five criteria in Table 4. Assign a score out of 2 for each criterion, and write one short comment giving the strengths of the design and one suggestion for improvement. A review submitted without a written comment earns no marks.

Submit your review through the Microsoft Form linked in the Lab 3 presentation on D2L. Reviews are accepted only through that form. Your written comment will be returned to the group whose design you reviewed, together with their grade.

You may not speak to the authors of the design you are reviewing, and you may not ask them for clarification. If a step is unclear, that is itself a finding — record it under Clarity of experiment.

You are reviewing the design, not the team. Write the comments you would want to receive: specific, technical, and directed at what would have to change. Your Section 7 mark in Lab 2 reflects both the quality of your own design and the quality of the review you provide.

[3] https://www.mathworks.com/help/control/ref/tf.html
[4] https://www.mathworks.com/help/control/ref/dynamicsystem.step.html

Table 4: Peer review criteria for the Lab 2 experiment design

Criteria	Description	Points
Clarity of Experiment (2 pts)	Objective and steps are clearly stated. Another team could execute the procedure exactly as written, without asking the authors a question.	
Parameter Identification (2 pts)	Correctly identifies the two motor parameters, and determines them independently rather than yielding only a combination of the two.	
Experimental Setup (2 pts)	Setup is practical, safe and logically organized. Voltage limits and any stall duration are explicitly stated.	
Data Collection & Method (2 pts)	Explains how each quantity will be measured and recorded, including the number of trials and the number of repetitions at each setting.	
Feasibility & Realism (2 pts)	Design is achievable with the available hardware and instrumentation, and leads to valid results.	
Total		**/ 10**

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